

stage1_standalone_repos_mapping

Standalone Repository Analysis Report

Executive Summary

All 8 required standalone repositories have been analyzed via GitHub API and partial clones. This document provides the structural and technical overview for each repository.

1. HelixAgent (github.com/HelixDevelopment/HelixAgent)

Property	Value
Language	Go + JavaScript (Playwright)
Go Version	Latest (from go.mod head)
Top-Level Items	150+ files/directories
Submodules	26+ declared in .gitmodules
Key Directory	cli_agents/ - 60+ agent integrations + 20+ analysis Markdown files
Special	Playwright dependency for browser automation

Submodules Declared

- containers (vasic-digital)
- Challenges (vasic-digital)
- EventBus, Concurrency, Observability
- Auth, Storage, Streaming, Security
- VectorDB, Embeddings, Database, Cache
- Messaging, Formatters, MCP_Module
- RAG, Memory, Optimization, Plugins
- Toolkit/SiliconFlow, Toolkit/Chutes
- LLMsVerifier (vasic-digital)
- External: modelcontextprotocol/servers

cli_agents Directory Contents

60+ CLI agent integration modules including analysis of: - Claude Code, Claude Squad - Aider, Cline, Codex - Plandex, GPT Engineer - Continue.dev, Cursor - And many more...

20+ analysis Markdown files documenting: - Feature comparisons - Architecture analysis - Porting strategies - API mappings

Key Insight

HelixAgent is the ANCESTOR project of HelixCode. The root AGENTS.md in HelixCode explicitly states: “Derived from HelixAgent AGENTS.md with HelixCode-specific enhancements”. HelixAgent’s `cli_agents` directory contains deep analysis of every major CLI agent in the ecosystem.

2. HelixLLM (github.com/HelixDevelopment/HelixLLM)

Property	Value
Language	Go
Go Version	1.26.1
Web Framework	Gin Gonic v1.12.0
Network	quic-go v0.59.0
Coverage	91% threshold
Submodules	43 via <code>replace</code> directives in <code>go.mod</code>
Operating Modes	6 (full, gateway, brain, knowledge, agents, control)

Architecture

- Single binary with 6 operating modes
- Modular LLM gateway with provider abstraction
- QUIC transport support
- Knowledge management backend
- Agent orchestration capabilities

Submodules (Partial List)

Config, EventBus, Observability, Concurrency, Auth, Storage, Streaming, Security, VectorDB, Embeddings, Database, Cache, Messaging, Formatters, MCP_Module, RAG, Memory, Optimization, Plugins, TOON, Watcher, I18n, RateLimiter, Document, Filesystem, HelixQA, Lazy, LLMOrchestrator, LLMProvider, Middleware, Planning, Recovery, ToolSchema, Agentic, BackgroundTasks, Containers, Conversation

3. HelixMemory (github.com/HelixDevelopment/HelixMemory)

Property	Value
Language	Go
Go Version	1.25
Backends	Mem0, Cognee, Letta, Graphiti

Property	Value
Interface	digital.vasic.memory

Architecture

- Drop-in memory replacement interface
- Multi-backend fusion architecture
- Graph-based memory (Graphiti)
- Structured memory (Cognee)
- Agent memory (Letta)
- Simple memory (Mem0)

4. HelixSpecifier (github.com/HelixDevelopment/HelixSpecifier)

Property	Value
Language	Go
Go Version	1.24.0
Packages	27 (21 core + 6 test suites)
Engine	3-pillar fusion

Core Components

1. **SpecKit** - Specification generation and parsing
2. **Superpowers** - Extended capability system
3. **GSD** (Get Shit Done) - Execution engine

5. helix_qa (github.com/HelixDevelopment/HelixQA)

Property	Value
Language	Go
Features	Docker, Kubernetes, OpenCV integration
Type	Testing framework with test banks
Top-Level Items	30+

Capabilities

- Automated QA session framework
 - Test bank management
 - Containerized testing
 - Visual regression (OpenCV)
 - Kubernetes test orchestration
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6. LLMsVerifier (github.com/vasic-digital/LLMsVerifier)

Property	Value
Language	Go
Provider Adapters	12
Protocol	ACP (Agent Communication Protocol)
Compression	Brotli
Deployment	Kubernetes manifests included
Top-Level Items	80+

Features

- Multi-provider LLM verification
 - ACP protocol implementation
 - Response compression
 - Benchmarking suite
 - Kubernetes-native deployment
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7. Challenges (github.com/vasic-digital/Challenges)

Property	Value
Language	Go
Type	Challenge/test framework
Architecture	Plugin-based

Purpose

- Generic challenge framework for testing AI systems
 - Plugin architecture for extensible test types
 - Used by HelixCode for capability verification
 - Test suite definitions and validators
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8. containers (github.com/vasic-digital/Containers)

Property	Value
Language	Go
Type	Container orchestration module
Backends	Docker, Podman, Kubernetes

Purpose

- Generic container abstraction layer

- Runtime-agnostic deployment
- Used by HelixCode and helix_qa for test isolation
- Infrastructure abstraction

Cross-Repository Relationship Matrix

Repository	Is Submodule Of	Has Submodules	Integrates With
HelixCode	-	87	All
HelixAgent	NOT IN HELIXCODE	26+	All (ancestor)
HelixLLM	NOT IN HELIXCODE	43 (replace)	HelixCode, HelixAgent
HelixMemory	NOT IN HELIXCODE	Unknown	HelixCode, HelixAgent
HelixSpecifier	NOT IN HELIXCODE	Unknown	HelixCode, HelixAgent
helix_qa	IN HELIXCODE	Unknown	HelixCode, Challenges, containers
LLMsVerifier	IN HELIXCODE	Unknown	HelixCode, HelixLLM
Challenges	IN HELIXCODE	Unknown	HelixCode, helix_qa
containers	IN HELIXCODE	Unknown	HelixCode, HelixQA, HelixAgent

Submodule Status Summary

Present in HelixCode (4/8)

- LLMsVerifier
- HelixQA
- Challenges
- Containers

Missing from HelixCode (4/8) - CRITICAL

- HelixAgent (ancestor project, 60+ cli_agents)
- HelixLLM (LLM gateway, 43 submodules)
- HelixMemory (memory fusion)
- HelixSpecifier (specification engine)

Integration Priority

1. **CRITICAL:** HelixAgent - Contains all CLI agent analysis and porting guides
2. **HIGH:** HelixLLM - Core LLM infrastructure with 43 submodules
3. **HIGH:** HelixMemory - Memory systems for context persistence
4. **HIGH:** HelixSpecifier - Specification and contract system
5. **MEDIUM:** challenges/containers/helix_qa/LLMsVerifier - Already present, need full initialization